



Inspiring Excellence

Assignment-01

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Course: CSE330

Section: 13

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Ans. to the ques. no-1

a

$$x = (3.165)_{10}$$

$$\begin{array}{r} 2 \overline{) 3} \\ \underline{2} \\ 1 \\ \underline{0} \\ 1 \end{array} \uparrow$$

$$\therefore (3.165)_{10} = (11.001010100)_2$$

$$\begin{array}{r} 0.165 \\ \times 2 \\ \hline 0 \cdot 33 \\ \times 2 \\ \hline 0 \cdot 66 \\ \times 2 \\ \hline 1 \cdot 32 \\ \times 2 \\ \hline 0 \cdot 64 \\ \times 2 \\ \hline 1 \cdot 28 \\ \times 2 \\ \hline 0 \cdot 56 \\ \times 2 \\ \hline 1 \cdot 12 \\ \times 2 \\ \hline 0 \cdot 24 \\ \times 2 \\ \hline 0 \cdot 48 \end{array}$$

6

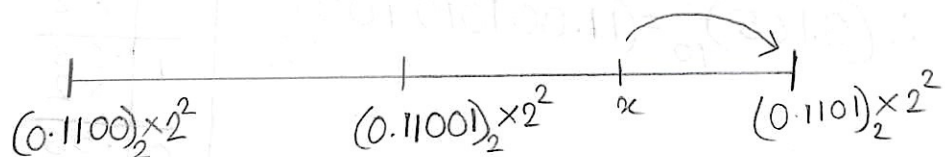
$$x = (3.165)_{10}$$

$$= (11.001010100)_2$$

$$= (0.\overbrace{11001010100}^{m=6})_2 \times 2^2$$

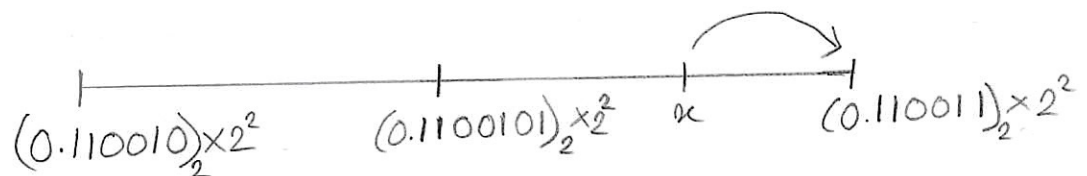
$$\quad \quad \quad \underbrace{\hspace{1.5cm}}_{m=4}$$

When $m = 4$,



$$\therefore fl(x) = (0.1101)_2 \times 2^2$$

When $m = 6$,



$$\therefore fl(x) = (0.110011)_2 \times 2^2$$

When $m = 4$,

$$\begin{aligned} fl(x) &= (0.1101)_2 \times 2^2 \\ &= (2^{-1} \times 1 + 2^{-2} \times 1 + 2^{-3} \times 0 + 2^{-4} \times 1) \times 2^2 \\ &= (3.25)_{10} \end{aligned}$$

$$\begin{aligned} \therefore \delta &= \frac{|fl(x) - x|}{|x|} \\ &= \frac{|3.25 - 3.165|}{|3.165|} \\ &= 0.026856 \text{ (approx.)} \end{aligned}$$

When $m = 6$,

$$\begin{aligned} fl(x) &= (0.110011)_2 \times 2^2 \\ &= (2^{-1} \times 1 + 2^{-2} \times 1 + 2^{-3} \times 0 + 2^{-4} \times 0 + 2^{-5} \times 1 + 2^{-6} \times 1) \times 2^2 \\ &= (3.1875)_{10} \end{aligned}$$

$$\begin{aligned} \therefore \delta &= \frac{|fl(x) - x|}{|x|} \\ &= \frac{|3.1875 - 3.165|}{|3.165|} \\ &= 7.109 \times 10^{-3} \text{ (approx.)} \end{aligned}$$

Ans. to the ques. no-2

a

For IEEE Normalised:

Without negative support:

$$\begin{aligned}\text{Maximum} &= (0.11111)_2 \times 2^1 \\ &= (2^{-1} \times 1 + 2^{-2} \times 1 + 2^{-3} \times 1 + 2^{-4} \times 1 + 2^{-5} \times 1) \times 2^1 \\ &= \left(\frac{31}{16}\right)_{10}\end{aligned}$$

$$\begin{aligned}\text{Minimum} &= (0.10000)_2 \times 2^{-1} \\ &= 2^{-1} \times 2^{-1} \\ &= \left(\frac{1}{4}\right)_{10}\end{aligned}$$

With negative support:

$$\text{Maximum} = (0.11111)_2 \times 2^1$$

$$= (2^{-1} \times 1 + 2^{-2} \times 1 + 2^{-3} \times 1 + 2^{-4} \times 1 + 2^{-5} \times 1) \times 2$$

$$= \left(\frac{31}{16}\right)_{10}$$

$$\text{Minimum} = -(0.11111)_2 \times 2^1$$

$$= -(\cancel{2^{-1} \times 1} + 2^{-2} \times 1 + 2^{-3} \times 1 + 2^{-4} \times 1 + 2^{-5} \times 1) \times 2$$

$$= -\left(\frac{31}{16}\right)_{10}$$

IEEE Denormalised form:

Without negative support:

$$\begin{aligned}\text{Maximum} &= (1.1111)_2 \times 2^1 \\ &= (2^0 \times 1 + 2^{-1} \times 1 + 2^{-2} \times 1 + 2^{-3} \times 1 + 2^{-4} \times 1) \times 2^1 \\ &= \left(\frac{31}{8}\right)_{10}\end{aligned}$$

$$\begin{aligned}\text{Minimum} &= (1.0000)_2 \times 2^{-1} \\ &= \left(\frac{1}{2}\right)_{10}\end{aligned}$$

With negative support:

$$\begin{aligned}\text{Maximum} &= (1.1111)_2 \times 2^1 \\ &= \left(\frac{31}{8}\right)_{10}\end{aligned}$$

$$\begin{aligned}\text{Minimum} &= -(1.1111)_2 \times 2^1 \\ &= -\left(\frac{31}{8}\right)_{10}\end{aligned}$$

b

Standard Form:

$$F = (0.d_1 d_2 d_3 d_4)_B \times B^e \\ = (0.1 d_2 d_3 d_4)_B \times B^e$$

$$\therefore \text{Combinations} = 2^3 = 8$$

$$\therefore \text{Total only (+ve) no.} = 2^3 \times 3 \\ = 24$$

$$\therefore \text{Total numbers both (+ve) \& (-ve)} = 2^3 \times 3 \times 2 \\ = 48$$

IEEE Normalised:

$$F = (0.1 d_1 d_2 d_3 d_4)_B \times B^e$$

$$\therefore \text{Combinations} = 2^4 = 16$$

$$\therefore \text{Total no. only (+ve)} = 2^4 \times 3 \\ = 48$$

$$\therefore \text{Total no. both (+ve) \& (-ve)} = 2^4 \times 3 \times 2 \\ = 96$$

IEEE Denormalised:

$$F = (1.d_1 d_2 d_3 d_4)_\beta \times \beta^e$$

$$\text{Combinations} = 2^4$$

$$\therefore \text{Total no. of (+ve)} = 2^4 \times 3 \\ = 48$$

$$\therefore \text{Total no. of both (+ve) \& (-ve)} = 2^4 \times 3 \times 2 \\ = 96$$

C

$$F = (0. d_1 d_2 d_3)_\beta \times \beta^e$$

For $e = -1$,

$$(0.100)_2 \times 2^{-1} = 2^{-1} \times 2^{-1} = \left(\frac{1}{4}\right)_{10}$$

$$(0.101)_2 \times 2^{-1} = (2^{-1} + 2^{-3}) \times 2^{-1} = \left(\frac{5}{16}\right)_{10}$$

$$(0.110)_2 \times 2^{-1} = (2^{-1} + 2^{-2}) \times 2^{-1} = \left(\frac{3}{8}\right)_{10}$$

$$(0.111)_2 \times 2^{-1} = (2^{-1} + 2^{-2} + 2^{-3}) \times 2^{-1} = \left(\frac{7}{16}\right)_{10}$$

For $e = 0$,

$$(0.100)_2 \times 2^0 = 2^{-1} \times 2^0 = \left(\frac{1}{2}\right)_{10}$$

$$(0.101)_2 \times 2^0 = (2^{-1} + 2^{-3}) \times 2^0 = \left(\frac{5}{8}\right)_{10}$$

$$(0.110)_2 \times 2^0 = (2^{-1} + 2^{-2}) \times 2^0 = \left(\frac{3}{4}\right)_{10}$$

$$(0.111)_2 \times 2^0 = (2^{-1} + 2^{-2} + 2^{-3}) \times 2^0 = \left(\frac{7}{8}\right)_{10}$$

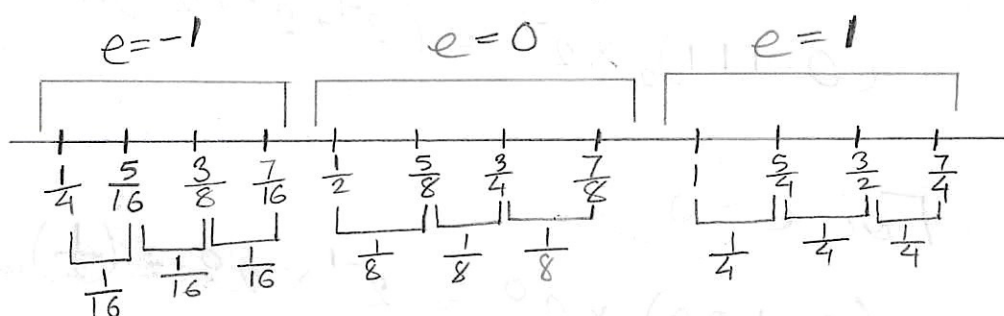
$$e=1,$$

$$(0.100)_2 \times 2^1 = 2^{-1} \times 2 = (1)_{10}$$

$$(0.101)_2 \times 2^1 = (2^{-1} + 2^{-3}) \times 2^1 = \left(\frac{5}{4}\right)_{10}$$

$$(0.110)_2 \times 2^1 = (2^{-1} + 2^{-2}) \times 2^1 = \left(\frac{3}{2}\right)_{10}$$

$$(0.111)_2 \times 2^1 = (2^{-1} + 2^{-2} + 2^{-3}) \times 2^1 = \left(\frac{7}{4}\right)_{10}$$



For specific exponent, the numbers are equally spaced. But for other values of e , numbers are not equally spaced. So, in a nutshell we can say the numbers are not equally spaced.